

Modulation of Release of Reactive Oxygen Species by the Contracting Diaphragm

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Recent reports have demonstrated that superoxide is released by the contracting diaphragm (1, 2). Moreover, extracellular scavengers of superoxide (i.e., exogenously administered superoxide dismutase) reduce diaphragm fatigue rate, arguing that superoxide released from contracting muscles may have functionally significant effects (1, 3). The mechanism by which free radical formation and release occurs has not, however, been determined, and all past studies of this phenomenon have been conducted at a single muscle length (the length of maximum force generation, L_0) and at a single level of carbon dioxide. The purpose of the present study was twofold: (1) to examine the effect of blockade of two free radical-generating pathways (i.e., to block cyclooxygenase with indomethacin and xanthine oxidase with oxypurinol) on superoxide release by the contracting diaphragm, and (2) to examine the effect of altering muscle length, carbon dioxide levels, and stimulation frequency on superoxide release during contraction. Studies were performed using an isolated, arterially perfused, rat diaphragm preparation in which superoxide release was assessed in real time by measuring arteriovenous cytochrome c reduction gradients across this muscle. We found that superoxide release during contraction was: (1) not altered by indomethacin administration, (2) partially reduced by oxypurinol administration, (3) reduced by decreasing muscle length, (4) reduced by increasing carbon dioxide concentrations, and (5) reduced by decreasing stimulation frequency. The first two findings indicate that xanthine oxidase pathways contribute to free radical formation under these circumstances but cyclooxygenase does not. The last three findings suggest that these common physiologic alterations have significant effects on free radical release by contracting muscle. **Stofan DA, Callahan LA, DiMarco AF, Nethery DE, Supinski GS. Modulation of release of reactive oxygen species by the contracting diaphragm.**

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Recent work has shown that oxygen-derived free radicals are generated in the diaphragm in response to strenuous contractions (1, 2). Moreover, administration of free radical scavengers to the contracting diaphragm acts to reduce the rate of development of diaphragm fatigue, suggesting that free radicals may play an important pathophysiologic role in the development of some forms of contraction-induced muscle dysfunction (3-6). Of note, reduction in diaphragm fatigue rate has been observed in response to administration of both low molecular weight scavengers (i.e., Tiron) (6) and high molecular weight cell-impermeant scavengers of either superoxide (i.e., superoxide dismutase) or its metabolic products (3-5). The response to these latter agents would seem to indicate that it is possible to use agents that scavenge extracellular free radicals in order to alter diaphragm fatigue rates. In keeping with this possibility, release of superoxide into the extracellular milieu by contracting muscle has been demonstrated in several reports (1, 2).

Much remains unknown, however, about contraction-induced free radical generation by the diaphragm. For one thing, the metabolic pathways responsible for the generation of free radicals in contracting skeletal muscle have not been firmly established. In addition, most of the studies examining free radical generation in the diaphragm and other skeletal muscles have been performed using experimental models (*in vitro* and *in situ* muscle preparations) in which physiologic variables have been tightly controlled and held within standard, "normal" limits, i.e., with muscles bathed in solutions having a "normal" P_{aCO_2} , with muscles held at a constant length at L_0 (the length at which the force generated in response to 1 Hz stimuli was maximal), etc. The clinical situations in which the diaphragm is loaded, however, are almost invariably associated with concomitant alterations in a number of systemic physiologic variables (e.g., alterations in arterial P_{aCO_2} levels, reductions in inspiratory muscle length). No previous study had been done, however, to examine the effect of alteration in any of these important physiologic variables on free radical release by the diaphragm.

The purpose of the present study was, therefore: (1) to examine the effect of blockade of two free radical-generating pathways (i.e., xanthine oxidase and cyclooxygenase pathways) on superoxide release by the contracting diaphragm, and (2) to examine the effect on superoxide release of three physiologic alterations (i.e., reductions in respiratory muscle length, increases in carbon dioxide concentrations, alterations

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in stimulation frequencies) that occur very commonly in clinical circumstances.

METHODS

Male adult Sprague-Dawley rats weighing 400 to 500 g were used in the present experiments. These animals were housed and fed according to ALAC guidelines at a Case Western Reserve University animal facility.

Isolated Diaphragm Preparation

A new diaphragm muscle perfusion technique was used for this experiment. This technique was developed to provide a simple means of directly infusing drugs and other agents into the arterial supply of the diaphragm while simultaneously collecting the venous effluent. This approach, which will be described in detail in the following paragraphs, was required because the present experiments call for serial determination of arteriovenous gradients of reduced cytochrome c concentrations across the diaphragm vascular bed. Although several previous investigators have measured diaphragm release or uptake of substances using retrograde perfusion of rodent diaphragms, retrograde perfusion results in achievement of nonphysiologically high capillary pressures in the diaphragmatic vascular bed (2, 7). In contrast, the present approach utilizing antegrade perfusion should result in physiologic arterial, capillary, and venous pressure in this muscle.

When preparing diaphragms in this fashion, rats were first anesthetized in a halothane chamber and then decapitated. The chest was then surgically opened and the thoracic aorta cannulated with a 16-gauge angiocatheter advanced to a level just proximal to the diaphragm. The aorta was then perfused using a nonrecirculating infusion system with oxygenated (95% O₂, 5% CO₂) Krebs-Henseleit solution at 22° C (pH, 7.4; NaCl, 135 mM; KCl, 5 mM; dextrose, 11.1 mM; CaCl₂, 2.5 mM; MgSO₄, 1 mM; NaHCO₃, 14.9 mM; NaHPO₄, 1 mM; insulin, 50 U/L). In all studies, we were able to initiate perfusion of the thoracic aorta with physiologic solution within 1 min of death. Arterial perfusion pressure was monitored using a Gould pressure transducer (Gould Inc., Instruments Division, Cleveland, OH) attached to the infusion system via a sidearm and maintained at 100 mm Hg throughout experiments. After establishing arterial perfusion, the abdomen was opened and the distal aorta ligated just below the origin of the inferior left phrenic artery. The liver, lungs, and heart were removed, and additional dissection was done to isolate the left hemidiaphragm, making sure not to disrupt the arterial supply to this portion of the diaphragm (i.e., the phrenic artery and intercostal arteries entering the left diaphragm). The right diaphragm and adjoining rib cage were removed after ligation of the intercostal arteries supplying these tissues. The remaining tissue (i.e., isolated, perfused left diaphragm strip, the contiguous ribs, spinal column, and aorta) was then secured to a rubber support with stainless steel pins. Dissection was used to reduce the left diaphragm to a muscle preparation that was approximately 2 to 2.5 cm in width and weighed 250 to 300 mg; when performing this dissection, care was taken to leave intact perfusion via the intercostal vessels and inferior phrenic artery. During this dissection, the sternal insertion of the muscle was removed and the muscle that remained consisted entirely of costal diaphragm. The central tendon end of the large diaphragm muscle strip was attached to a rigid metal rod connected, in turn, to a force transducer (Grass FT 10; Grass Instruments, Quincy, MA) positioned directly above this muscle. To permit venous effluent sampling from this muscle, a siphon (consisting of a piece of polyethylene tubing) was inserted into the left phrenic vein draining it. To minimize the development of excessive venous pressures in this drainage system, the distal end of the siphon was placed 4 cm below the level of the inferior margin of the diaphragm. Fine bipolar stainless steel stimulating electrodes were next inserted into the muscle strip. Diaphragm length was set to Lo (the length at which the force generated in response to a 1 Hz stimuli was maximal); this length was measured by the use of a calibrated graticule. We should note that preliminary studies have indicated that this diaphragm preparation behaves in an isotropic fashion, with all portions of the muscle achieving maximal force simultaneously as the muscle is stretched along its long axis. As a result, we assessed muscle length at the midportion of the diaphragm (i.e., halfway along its short axis) when adjusting muscle length.

The diaphragm muscle preparation was then covered with polyethylene wrap and a 30-min equilibrium period was provided before beginning the experimental protocol. At the beginning of this 30-min period, the perfusion solution was changed to Krebs-Henseleit solution containing cytochrome c (*see below*).

Cytochrome c Assay

To assess superoxide release by the contracting isolated perfused diaphragm, a modification of the techniques described by Reid and colleagues (1) and Kolbeck and coworkers (2) was employed. This approach takes advantage of the fact that superoxide reacts with and reduces cytochrome c. As a result, measurement of superoxide dismutase-suppressible cytochrome c reduction provides a means of assessing superoxide formation. For the purposes of the present experiment, we infused cytochrome c containing solution (i.e., Krebs-Henseleit at pH 7.4 containing cytochrome c, 10 μ M) and measured the rate at which the infused cytochrome c was reduced. To minimize temperature-related autoreduction or photoreduction of cytochrome c, these experiments were performed in a darkened laboratory, the infusion apparatus was foil-wrapped, and both the infusion apparatus and the muscle preparation were kept at 22° C. Once cytochrome c infusion was initiated, venous effluent from diaphragm preparations was collected in vials changed at 5-min intervals. The absorbance of each venous sample was measured spectrophotometrically at 550 nm (Shimadzu Model UV 1201) and the volume of each sample was recorded. The absorbance of the arterial perfusate was also determined at 5-min intervals. The rate of cytochrome c reduction by the diaphragm was calculated by multiplying the difference between arterial and venous absorbance times venous sample volume flow rate (i.e., in ml/min) and the extinction coefficient for cytochrome c reduction (29,500/mole).

When performing these experiments, it was important to make sure that reduction of cytochrome c was not substrate-limited (i.e., this would occur if insufficient oxidized cytochrome c was provided in the arterial perfusate, limiting the total amount of cytochrome c that could be reduced per unit time). For this reason, we incubated a sample of the arterial reservoir with a superoxide-generating solution (i.e., xanthine oxidase/hypoxanthine) on each day of study to determine the maximum cytochrome c signal that was possible (i.e., the maximum cytochrome c reduction per unit volume). For this assessment we combined 0.02 U of xanthine oxidase with 5 mM hypoxanthine and 10 μ M cytochrome c in a 1-ml quartz cuvette; this mixture was then placed in a spectrophotometer and the amount of reduced cytochrome c was assessed from the absorbance at a wavelength of 550 nm. This reaction was followed to completion, i.e., as judged by a plateau in the absorbance value. We always found that the highest signal measured each day (i.e., the maximum observed amount of cytochrome c reduced in a unit volume) was 2- to 3-fold lower than the capacity of our perfusate for reduction.

We also incubated each of the substances used in these experiments with oxidized cytochrome c *in vitro* to determine if any of these agents directly reduces cytochrome c. We found that oxypurinol, indomethacin, and elevated CO₂ concentrations did not elicit *in vitro* reduction of cytochrome c. Finally, to verify that the cytochrome c reduction signal obtained during diaphragmatic contraction was due to superoxide elaboration and not to reaction of cytochrome c with some other metabolite, we assessed the effect of superoxide dismutase (SOD) administration on cytochrome c reduction rate by the contracting diaphragm. For this work, one group of diaphragm preparations (n = 5) was infused with Krebs-Henseleit containing cytochrome c, whereas SOD (4,000 U/ml) was added to the solution infused into the second group of studies (n = 5). In both groups of studies, we first collected baseline measurements of cytochrome c reduction rate by the diaphragm for 15 min. Diaphragms were then electrically stimulated to repetitively contract for 10 min (trains of 20 Hz stimuli with a train duration of 500 ms, at a train rate of 15/min). Assessment of diaphragm cytochrome c reduction rate was continued throughout the 10 min contraction trial and for an additional 10-min period postcontraction.

Experimental Protocols

Two groups of studies were performed. Group A studies were designed to evaluate the effect of blockade of two pathways of free radical generation (i.e., xanthine oxidase and cyclooxygenase) on super-

oxide release by the contracting diaphragm, and Group B studies examined the effect of changing muscle length, CO₂ concentrations, and stimulation frequency on superoxide release. A similar experimental design was followed for all experimental groups: (1) diaphragm preparations were set up, (2) a 30-min equilibration period was provided during which time preparations were infused with physiologic solution containing cytochrome c, (3) paired arteriovenous samples were obtained at 5-min intervals over a 15-min period and used to calculate baseline diaphragmatic cytochrome c reduction gradients, (4) a 10-min period of rhythmic contraction was provided during which time assessment of diaphragm cytochrome c reduction rates was continued at 5-min intervals, (5) additional assessment of cytochrome c reduction rates was made over the first 10 min after cessation of contraction, and (6) diaphragms were removed and weighed. The basic stimulation paradigm employed during the 10-min contraction trial for these experiments was as follows: two twitches were evoked, a force-frequency curve was then constructed by sequentially stimulating muscles with trains of 1, 10, 20, 50, and 100 Hz stimuli (train duration, 800 ms; trains separated by 5 s) and this was immediately followed by 10 min of repetitive contraction in response to trains of 20 Hz stimuli (train duration, 500 ms; train rate, 15/min).

The standard conditions employed for these experiments were with muscle length set to Lo, with infusing solutions equilibrated to gas containing 5% CO₂/95% O₂, and with stimulus trains during the repetitive contraction trial set to a 20-Hz pulse frequency. Deviations from these standard conditions for specific experiments are provided below.

For Group A experiments, two comparisons were made: (1) we compared cytochrome c reduction rates over time between oxypurinol-treated and saline-treated control animals, and (2) comparison was made of cytochrome c reduction rates between indomethacin-treated and saline-treated control animals. For the set of oxypurinol studies, animals were injected intraperitoneally with oxypurinol 50 mg/kg 12 h prior to the death of the animal and additional oxypurinol (50 mg/L) was added to the diaphragmatic perfusate used during *in vitro* experimentation. Control animals for this study received an intraperitoneal injection of saline 12 h prior to their deaths. For the indomethacin studies, this agent was added (10 mg/ml) to the aortic infusion used during *in vitro* experimentation. The concentrations of oxypurinol and indomethacin used were based on previous work demonstrating the effect of these agents at these doses to adequately inhibit the targeted enzyme systems under *in vivo* conditions (11, 16).

For Group B studies, three comparisons were made: (1) comparison of cytochrome c reduction rates between diaphragms studied at a length of Lo and diaphragms studied at 65% of Lo, (2) comparison of diaphragms perfused with physiologic solution bubbled with 5% CO₂ and muscles perfused with solution equilibrated with 15% O₂, and (3) comparison of diaphragms stimulated with 20 Hz trains during the repetitive contraction trials and muscles stimulated with 5-Hz stimulus trains during these trials. For the set of muscles studied at 65% Lo, muscle length was reduced to this length at the beginning of the 30-min equilibration period provided immediately after diaphragm preparation. For the set of experiments studied with a high carbon dioxide concentration, infusion of Krebs-Henseleit solution containing cytochrome c and bubbled with gas containing 15% CO₂, 85% O₂ was initiated at the beginning of the equilibration period. For both high (15%) and low (5%) CO₂ sets, pH of infusing solutions was adjusted to 7.4, i.e., sufficient NaOH was added to the high CO₂ solution to bring pH to a normal level.

Data Analysis

Diaphragm force was expressed as normalized for cross-sectional area according to the following formula: Force/cm² = Force × Muscle length × 1.06 / Muscle weight.

RESULTS

Effect of SOD on Diaphragm Cytochrome c Reduction Rates

Diaphragm cytochrome c reduction rates (CCRR) over time before, during, and after a repetitive stimulation trial is shown in Figure 1, *bottom panel*. The *top panel* of this same figure displays force generation over time for these same experiments. Prior to contraction trials, CCRR was low and stable. Repetitive contraction was associated with an immediate in-

crease in CCRR, which rose to approximately three times the basal rate over the first 5 min of the contraction. Diaphragm force generation and CCRR decreased over the second 5 min of the contraction trial, and CCRR subsequently fell to basal levels after cessation of contraction ($p < 0.001$ for comparison baseline CCRR to CCRR over the first 5 min of the contraction trial).

SOD administration did not significantly alter the shape of the diaphragm force-frequency curve (Figure 2, *panel A*) and did not change the absolute force generated by the diaphragm at 50 Hz (i.e., force generation in response to 50 Hz; stimulation was 24.4 ± 2.0 N/cm² for control and 24.9 ± 2.4 N/cm² for SOD-treated muscles). Administration of SOD, however, resulted in near-complete suppression of CCRR and was also associated with a small increase in the level of diaphragm force generated over time during repetitive contraction trials ($p < 0.01$ for comparison of CCRR during the first 5 min of contraction with and without SOD administration). The fact that SOD suppressed contraction-related CCRR is consistent with previous reports and indicates that contraction-related CCRR under these conditions is due to superoxide elaboration by the diaphragm. Failure of SOD to ablate basal CCRR (i.e., CCRR by resting muscle) would indicate, however, that some other molecular species is responsible for CCRR in resting muscle.

Diaphragm venous flow was not significantly different between the SOD experiments and controls. Specifically, diaphragm venous effluent flow rate was, 0.38 ± 0.08 , 0.40 ± 0.04 , and 0.37 ± 0.09 ml/min, respectively, at the beginning of data collection, over the first 5 min of the contraction period, and at the end of the recovery period for control studies and was

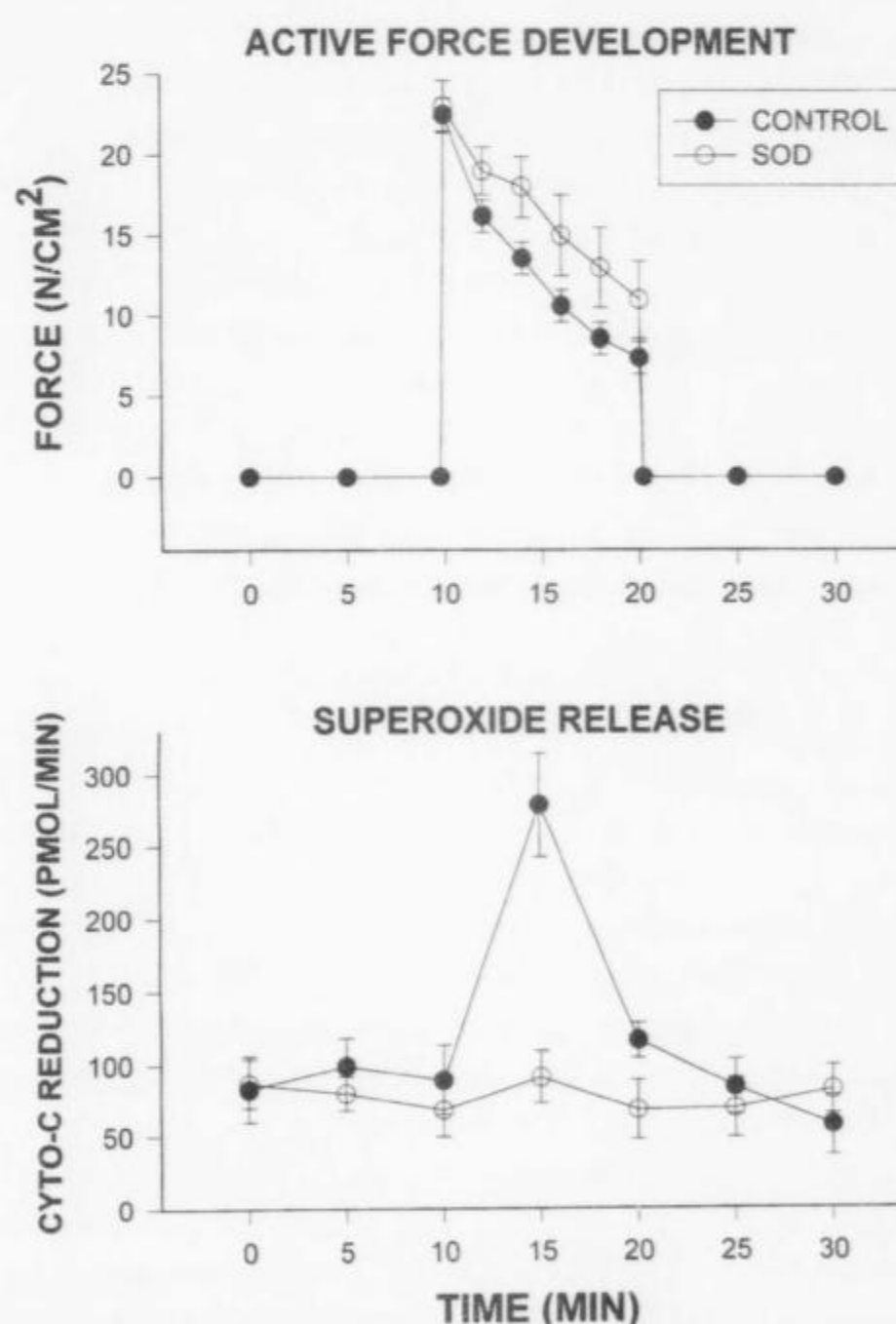


Figure 1. Active force development (*top panel*) and diaphragm superoxide release (*bottom panel*) as a function of time for control ($n = 5$) and superoxide dismutase (SOD) ($n = 5$) treated preparations. Preparations were electrically stimulated to undergo rhythmic contractions between 10 and 20 min. Cytochrome c reduction rate (y-axis on *bottom panel*) was taken as an index of superoxide release. Error bars represent 1 SE.

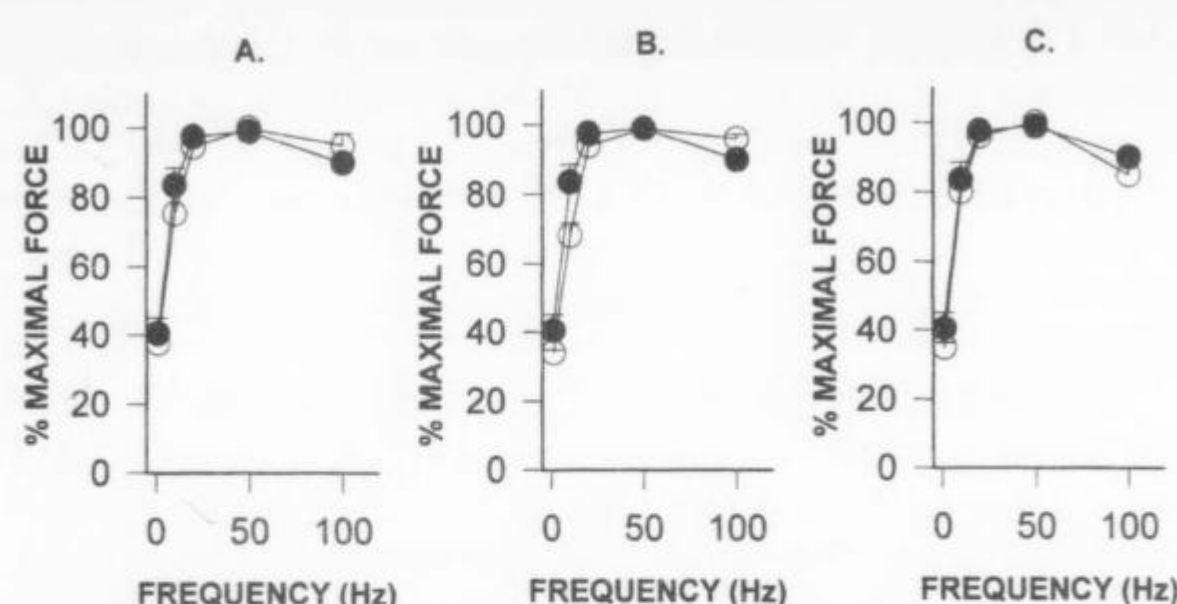


Figure 2. Relationship of force output as a function of frequency of stimulation to diaphragm preparations. From left to right are shown data for: (A) comparisons of diaphragms studied with (open circles) and without (closed circles) SOD administration; (B) comparisons of diaphragms studied with (open circles) and without (closed circles) indomethacin administration, and (C) comparisons of diaphragms studied with (open circles) and without (closed circles) oxypurinol administration.

0.41 ± 0.14 , 0.59 ± 0.20 , and 0.39 ± 0.09 ml/min at these same time points for experiments after SOD administration (arterial perfusion pressure was 100 mm Hg throughout studies in both groups of experiments).

Group A Studies: Effect of Indomethacin and Oxypurinol on Diaphragm CCRR

A comparison of data from studies performed with and without indomethacin administration is shown in Figures 2 and 3 (Figure 2, panel B, displays force frequency curves with and without indomethacin; Figure 3 shows CCRR over time for indomethacin experiments). Indomethacin neither altered the shape of the force-frequency curve of the diaphragm nor affected the force generated in response to 50 Hz stimulation, which averaged 24.4 ± 2.0 N/cm² for control and 23.2 ± 0.9 N/cm² for indomethacin experiments. Indomethacin administration was, however, associated with a small downward shift in the baseline level of CCRR by the diaphragm (i.e., CCRR by resting muscle). The magnitude of the increment in CCRR (Figure 3) induced by contraction was, however, similar in control and indomethacin groups, indicating that this agent does not alter the contraction-related rise in CCRR. Indomethacin had no effect on the rate of fall over force over time during repetitive contraction (Figure 3, top panel) ($n = 5$). Diaphragm venous flow was not altered by indomethacin, averaging 0.38 ± 0.08 , 0.40 ± 0.04 , and 0.37 ± 0.09 ml/min, respectively, at the beginning of data collection, over the first 5 min of the contraction period, and at the end of the recovery period for control studies and was 0.39 ± 0.06 , 0.50 ± 0.10 , and 0.55 ± 0.10 at these same time points for indomethacin experiments (arterial perfusion pressure was 100 mm Hg throughout studies in both groups of experiments).

In contrast to indomethacin, oxypurinol did not alter baseline CCRR, but it blunted the magnitude of the contraction-related increase in CCRR in the diaphragm preparation ($p < 0.05$) (Figure 4). Oxypurinol had no effect on the baseline diaphragm force-frequency relationship (Figure 2, panel C), but it did slightly reduce the rate of fall of force over time during the repetitive contraction trial (Figure 4, top panel). Diaphragm venous flow was not altered by oxypurinol and averaged 0.38 ± 0.08 , 0.40 ± 0.04 , and 0.37 ± 0.09 ml/min, respectively, at the beginning of data collection, over the first 5 min of the contraction period, and at the end of the recovery pe-

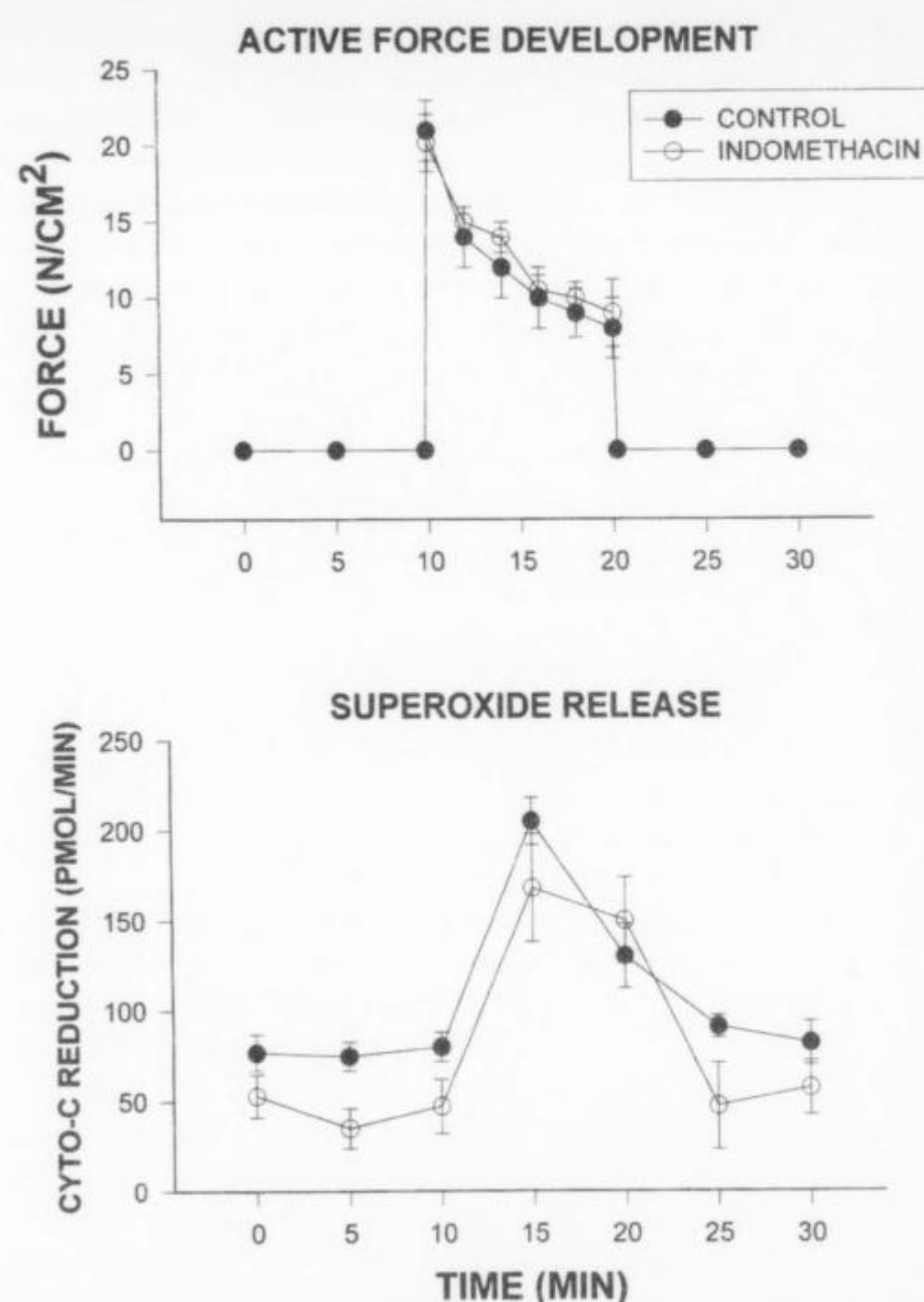


Figure 3. Active force (top panel) and superoxide release (bottom panel) over time for control ($n = 5$) and indomethacin ($n = 5$) treated diaphragm preparations. Rhythmic contractions were electrically induced between 10 and 20 min.

riod for control studies; venous flow was 0.46 ± 0.10 , 0.42 ± 0.07 , and 0.52 ± 0.2 ml/min at these same time points for oxypurinol experiments.

Group B Studies: Effect of Alterations in Muscle Length, Carbon Dioxide Concentrations, and Stimulation Frequency on Diaphragm CCRR

As expected, reducing muscle length was associated with a reduction in the level of force generated during force-frequency curve determination and during repetitive contraction trials (Figures 5, panel A, and Figure 6, top panel). More importantly, reductions in length were also accompanied by a large reduction in contraction-related increases in CCRR (Figure 6, bottom panel) ($p < 0.05$ for comparison of CCRR during the first 5 min of contraction between muscles held at Lo and at 65% Lo). Diaphragm venous flow was not significantly altered by reducing muscle length in these experiments. Specifically, venous flow was 0.38 ± 0.1 , 0.40 ± 0.1 , and 0.37 ± 0.1 ml/min, respectively, at the beginning of data collection, over the first 5 min of the contraction period, and at the end of the recovery period for control studies and was 0.78 ± 0.09 , 0.58 ± 0.13 , and 0.72 ± 0.1 ml/min at these same time points for experiments performed at a shortened muscle length ($p < 0.03$ for comparison of flow rates between the two groups at the beginning of data collection and at the end of the recovery period).

Increases in perfusate carbon dioxide levels did not alter the shape of the diaphragm force-frequency relationship (Figure 5, panel B) and did not alter the absolute force generated in response to 50 Hz electrical stimulation (this averaged 24.4 ± 2.0 N/cm² in control and 22.5 ± 2.0 N/cm² in high carbon dioxide experiments). An increase in carbon dioxide con-

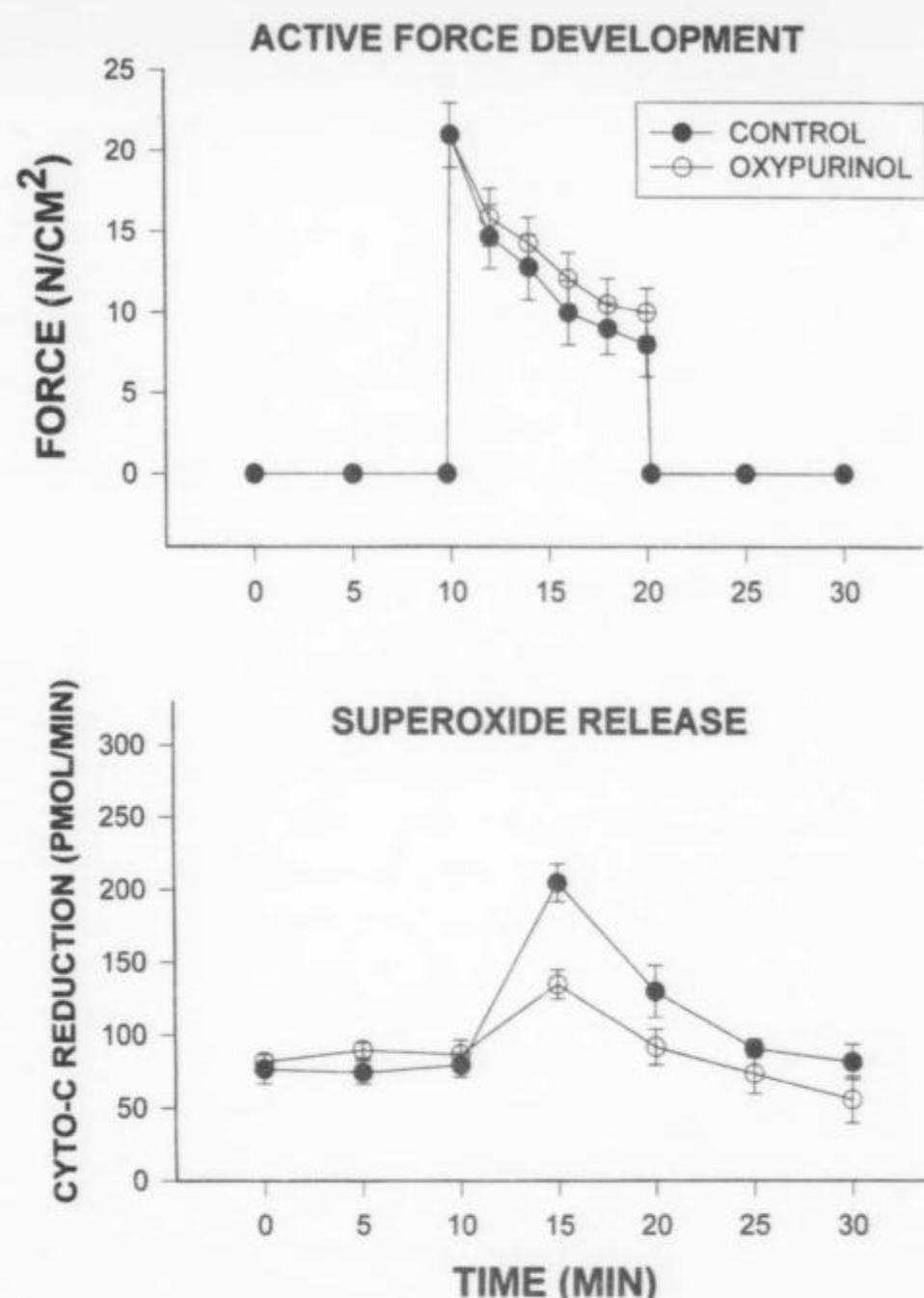


Figure 4. Active force (top panel) and superoxide release (bottom panel) over time for control ($n = 5$) and oxypurinol ($n = 5$) treated diaphragm preparations. Rhythmic contractions were electrically induced between 10 and 20 min.

centration was, however, associated with a decrease in diaphragm muscle CCRR, as shown in the bottom panel of Figure 7 ($p < 0.02$ for comparison of CCRR during contraction for 5% and 15% CO_2 groups). Elevations in carbon dioxide concentrations did not have a significant effect on the rate of fall of force over time during repetitive contraction trials (Figure 7, top panel). Diaphragm venous flow was not altered by increases in carbon dioxide levels in these experiments; specifically, venous flow was 0.38 ± 0.1 , 0.40 ± 0.1 , and 0.37 ± 0.1 ml/min, respectively, at the beginning of data collection, over the first 5 min of the contraction period, and at the end of

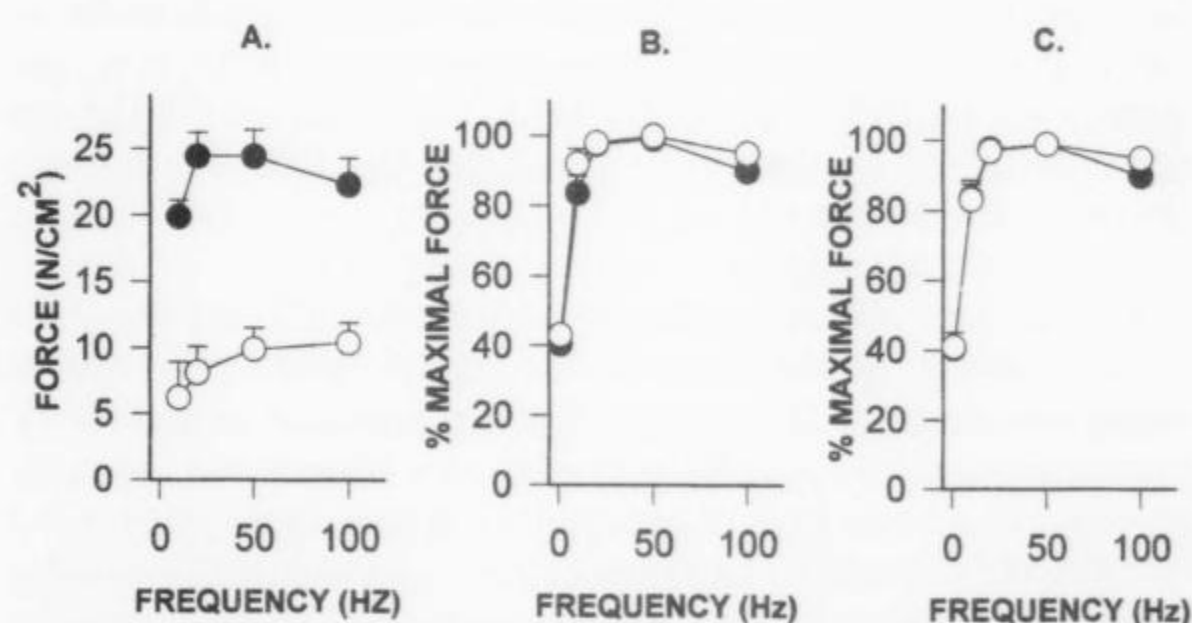


Figure 5. Relationship of force output as a function of frequency of stimulation to diaphragm preparations. From left to right are shown data for: (A) comparisons of diaphragms studied with muscle length adjusted to Lo (closed circles) and muscles length adjusted to 65% of Lo (open circles); (B) comparisons of diaphragms studied with 5% CO_2 (closed circles) and 15% CO_2 (open circles), and (C) comparisons of diaphragms studied with repetitive contraction trials of 20 Hz (closed circles) and 5 Hz (open circles).

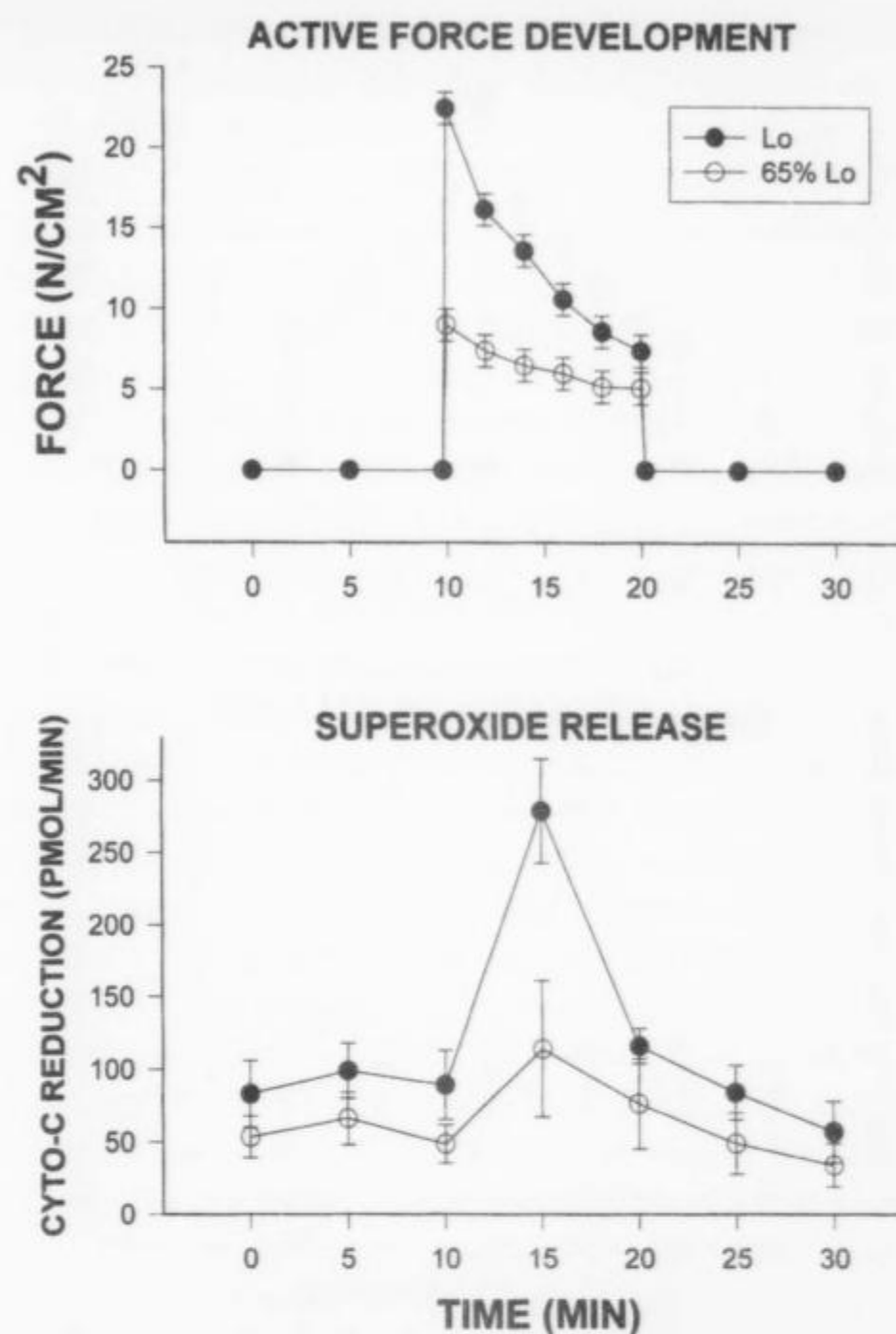


Figure 6. Active force (top panel) and superoxide release (bottom panel) over time for diaphragm preparations studied at muscle length of Lo ($n = 5$) and at a muscle length that was 65% Lo ($n = 5$). Rhythmic contractions were electrically induced between 10 and 20 min.

the recovery period for control studies (i.e., 5% CO_2) and was 0.40 ± 0.10 , 0.40 ± 0.10 , and 0.40 ± 0.12 ml/min at these same time points for experiments performed with 15% CO_2 .

Reductions in the frequency used for stimulation trains during repetitive contraction trials (i.e., trains of 20 Hz impulses were used for one set of these experiments, whereas 5 Hz trains were used for the second set) resulted, as expected, in reductions in the forces generated during repetitive contraction trials (see Figure 8, top panel); force frequency curves for this group of studies is shown in Figure 5, panel C. Reduction in train stimulation frequency was associated with a reduction in contraction-related increases in CCRR across the diaphragm (Figure 8, bottom panel) ($p < 0.01$ for comparison of CCRR during the first 5 min of contraction trials for 5-Hz and 20-Hz experimental groups). Diaphragm venous flow was similar for studies done with 5-Hz and 20-Hz electrical stimulation during repetitive contraction trials. Specifically, venous flow was 0.33 ± 0.04 , 0.40 ± 0.07 , and 0.33 ± 0.05 ml/min, respectively, at the beginning of data collection, over the first 5 min of the contraction period, and at the end of the recovery period for 5-Hz studies and was 0.38 ± 0.1 , 0.40 ± 0.1 and 0.37 ± 0.1 ml/min at these same time points for 20-Hz experiments.

DISCUSSION

Methodologic Issues

We measured cytochrome c reduction rates across the diaphragm and took this as an index of release of reactive oxygen species derived from superoxide in the current study. This interpretation is supported by the fact that administration of SOD, a very specific superoxide scavenger, ablated the contraction-associated increase in cytochrome c reduction by the

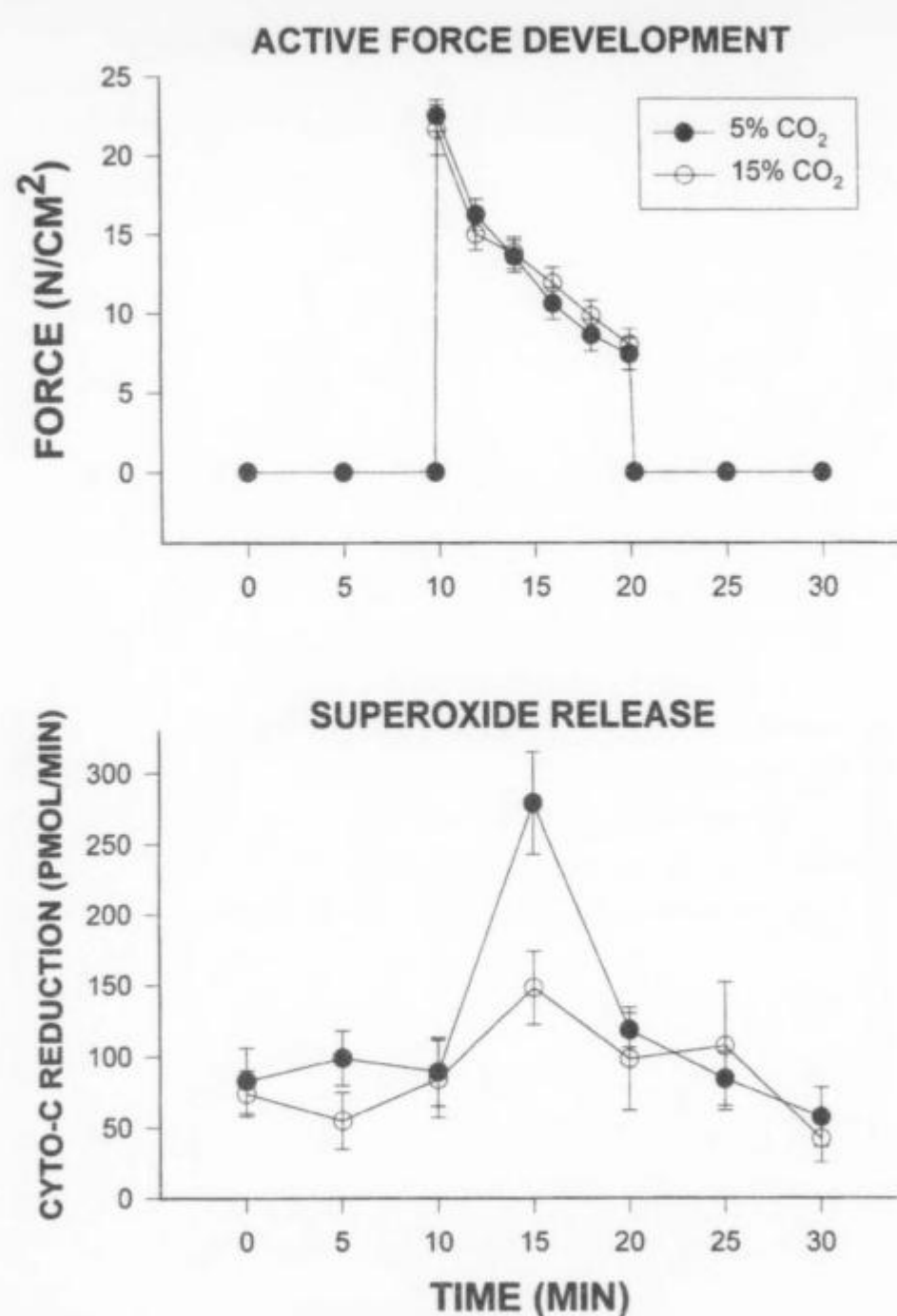


Figure 7. Active force (top panel) and superoxide release (bottom panel) over time for diaphragm preparations studied with a bath carbon dioxide (CO₂) concentration of 5% (n = 5) and at a CO₂ concentration of 15% (n = 5). Rhythmic contractions were electrically induced between 10 and 20 min.

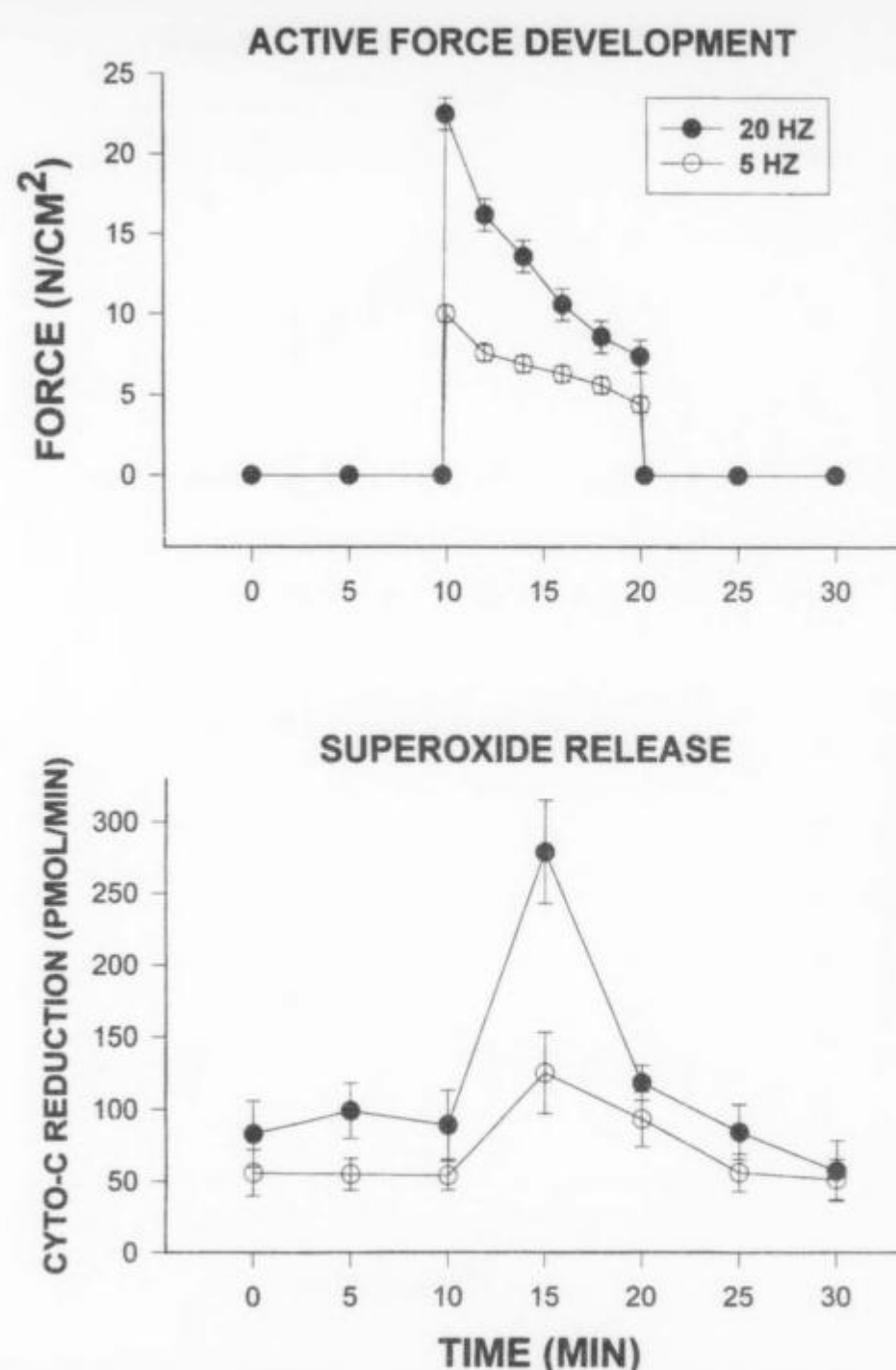


Figure 8. Active force (top panel) and superoxide release (bottom panel) over time. Rhythmic contractions were electrically induced between 10 and 20 min; one group of preparations was electrically stimulated with trains of 20 Hz stimuli (n = 5), whereas the other group was stimulated with trains of 5 Hz stimuli (n = 5).

diaphragm, as shown in Figure 1. This approach is also consistent with several recent reports that have also used cytochrome c reduction as an index of free radical release by the diaphragm. We think it most likely that the source of the superoxide being measured in this study is the diaphragm myocytes *per se* since there is a marked increase in cytochrome c reduction with a stress (muscle contraction) that would have its greatest metabolic effects on myocytes. We cannot exclude the possibility, however, that some or even all of the reactive oxygen species responsible for cytochrome c reduction in this experiment were derived from diaphragm endothelium. Muscle contraction would be expected to alter the shear stresses applied to the vascular bed of the diaphragm, and it is possible that this latter stimulus could induce endothelial superoxide formation and release. There is no way of distinguishing these two possible sources of reactive species using the current technique, and exact identification of the source of the free radical species being generated by the approach used in this study will require additional experimentation. Regardless of the cell of origin of the free radical species being measured in this study, however, our observations suggest that this phenomenon has an important relationship to diaphragm force-generation during contraction.

We performed these studies at a temperature of 22° C because higher temperatures artifactually alter cytochrome c reduction. Use of this low temperature may, however, have depressed superoxide generation below that present at higher, more physiologically relevant temperatures (i.e., 37° C). In addition, since these studies were not performed using an immersed preparation, curarization of the diaphragm and field stimulation to activate muscle fibers was not possible. As a re-

sult, transmission failure (i.e., neuromuscular fatigue) may have contributed to the fall in force generation seen during repetitive contraction trials. This fact would not be expected, however, to obviate any of the findings reported in the study.

The only intervention examined in this study that appreciably altered diaphragm perfusion was a reduction in muscle length (flow was higher when muscle length was short). Moreover, flow was relatively constant over time in these experiments, suggesting a constant level of muscle vasodilatation throughout experimental trials. As a result, the time-dependent alteration in CCRR observed during these experiments (i.e., increases in CCRR with contraction, inhibition of CCRR in response to SOD infusion) were not due to variations in muscle vascular flow rates.

This study attempted to survey a large number of conditions and interventions (administration of SOD, indomethacin, oxypurinol, alterations in muscle length, variation in carbon dioxide concentrations, alterations in frequency of stimulation) to determine if any of these perturbations altered our index of superoxide release by the contracting diaphragm. When performing these studies, we examined relatively extreme conditions, e.g., raising carbon dioxide levels to the highest range seen clinically, reducing muscle length to the shortest lengths likely to occur in clinical pathophysiologic situations, examining relatively high concentrations of xanthine oxidase and cyclooxygenase inhibitors. The utility of this approach and these studies is that we were able to identify some issues that can now be dismissed as being unimportant (i.e., the role of the cyclooxygenase pathway in mediating superoxide release by the diaphragm) and others that strikingly alter diaphragm super-

oxide release (e.g., increasing carbon dioxide concentrations). Future investigations will be needed to more precisely identify the mechanisms of these alterations and to more precisely calibrate the effect of these interventions on muscle free radical formation.

Free Radicals and Diaphragm Fatigue

A number of previous studies have provided support for the hypothesis that muscle contraction is associated with an increase in free radical formation by skeletal muscle and that this increased radical formation plays a role in the development of muscle fatigue (1–6, 8, 20). For example, several reports have detected an increase in markers of superoxide, hydrogen peroxide and hydroxyl radical formation in contracting skeletal muscle, including evidence of alterations in the EPR and ESR spectra, formation of hydroxysalicylate in muscles superfused with salicylate, and an increase in markers of lipid peroxidation and protein oxidation (1, 2, 9, 10). Other studies have found that administration of several different free radical scavengers (SOD, PEG-SOD, catalase, DMSO, N-acetylcysteine, Tiron) to muscle prior to and during contraction reduces the rate of development of muscle fatigue, arguing that free radicals formed during contraction interact with and alter the subcellular elements responsible for force generation (1–6). In one of these latter studies, Reid and colleagues (1) found that addition of SOD to the medium bathing *in vitro* diaphragm muscle strips resulted in a reduction in the rate of fall of force over time. In this same study, this group found that contracting diaphragm reduced cytochrome c added to the bathing medium, and that this reduction was ablated by coadministration of SOD. This combination of findings was taken as evidence for superoxide release into the extracellular medium. Qualitatively similar findings have been obtained by Kolbeck and colleagues (2), who found that diaphragmatic contraction resulted in a reduction of cytochrome c contained in a physiologic solution used to perfuse this muscle.

The observation that SOD administration reduces the fatigue of electrically stimulated muscle bundles is of some importance. SOD is a large, high molecular weight molecule that is thought to be largely confined to the extracellular space after administration. As such, this enzyme should primarily, if not entirely, act to scavenge extracellular superoxide species. The fact that administration of this agent affects muscle function (e.g., it alters the development of muscle fatigue) would seem to imply, therefore, that extracellular superoxide release by contracting muscle has important physiologic effects. If this line of reasoning is correct, conditions that increase extracellular superoxide concentrations should increase fatigue, whereas factors (like SOD administration) that reduce extracellular superoxide levels should decrease fatigue.

Effect of Xanthine Oxidase and Cyclooxygenase Inhibitors on Superoxide Release during Diaphragmatic Contraction

The present study was designed in part to extend the observations made by Reid and colleagues (1) and Kolbeck and coworkers (2) by determining if superoxide release from the contracting diaphragm could be ablated by administration of inhibitors of either xanthine oxidase (i.e., oxypurinol) or cyclooxygenase. Like Reid and colleagues (1), we took cytochrome c reduction as an indicator of superoxide release by the contracting diaphragm, and verified the fact that addition of SOD to our preparation ablated cytochrome c reduction. We subsequently found that cyclooxygenase inhibitor administration had no effect on superoxide release from the diaphragm, arguing that contraction-related free radical release by the diaphragm is not dependent upon the activity of the cy-

clooxygenase pathway. Indomethacin, the cyclooxygenase inhibitor tested in the current experiment, has previously been shown to reduce vascular abnormalities in two models of brain injury in intact cats (11, 12). This protective effect of indomethacin on the brain, moreover, was demonstrated in this previous study, to be linked to an action of this agent to inhibit the generation of free radicals associated with cyclooxygenase-mediated prostaglandin synthesis (11).

On the other hand, we found that administration of oxypurinol partially reduced contraction-related superoxide release from the diaphragm. This finding is consistent with a previous report in dog limb skeletal muscle, which found that allopurinol administration altered muscle fatigue rate and concluded, from that observation, that xanthine oxidase-derived free radicals may play a role in the development of muscle fatigue (13). In contrast, however, two recent reports were unable to demonstrate a significant effect of either allopurinol or oxypurinol on the development of diaphragmatic fatigue or lipid peroxidation during respiratory loading of intact rats (14, 15).

The most likely explanation for these discrepant results regarding the effect of xanthine oxidase inhibitors on muscle free radical generation is that more than one generating system may contribute to formation of free radicals during contraction in muscle, and that the relative contribution of different generating systems may vary depending upon animal species and the particular type of exercise being examined (19, 21). Supporting this possibility, superoxide release from the contracting diaphragm was not completely suppressed by oxypurinol administration in the present study, even though this agent was administered in adequate concentrations, based on past work, to have completely inhibited muscle xanthine oxidase activity (16). Also in keeping with this concept, multiple free radical-generating pathways have been reported to be involved in mediating free radical formation in several other tissues, including vascular endothelium after ischemia-reperfusion, and in lung after hyperoxic stress (17–19, 21). In vascular endothelium, for example, intracellular free radical formation appears to be inhibitable by administration of xanthine oxidase, but cell release of reactive oxygen species is not (17). Although xanthine oxidase pathways may not, therefore, contribute to free radical generation under all circumstances, the present data make a strong argument that this pathway can elicit appreciable free radical formation in contracting skeletal muscle under some conditions.

Effect of Alterations in Muscle Length, Stimulation Frequency, and Carbon Dioxide Concentrations on Muscle Release of Superoxide during Contraction

The second goal of these studies was to determine the effect of several important physiologic alterations on free radical release by the isolated diaphragm. We chose, specifically, to examine the effect of altering muscle length, muscle stimulation frequency, and arterial carbon dioxide concentrations on superoxide release by the *in vitro* contracting diaphragm. These particular physiologic alterations were studied because this group of parameters is known to change radically in patients with lung disease, the primary pathophysiologic state in which free radical-mediated muscle dysfunction may be a clinical problem.

The rationale for examining free radical generation by the diaphragm is related to the fact that the inspiratory muscles are commonly shortened during the development of respiratory failure in several disease states (asthma, exacerbations of chronic obstructive lung disease, bronchitis, and other disease processes that induce lung hyperinflation) (22). The present findings would argue that such hyperinflation-induced inspira-

tory muscle shortening per se (i.e., providing other factors, including stimulation frequency, remain constant) should act to decrease muscle free radical release. We also found that reductions in stimulation frequency from 20 Hz to 5 Hz acted to reduce superoxide release by the contracting diaphragm. This observation would, like the findings from our length experiment, tend to support the concept that free radical release is linked, to some degree, to the average force generated during a series of contractions (or the metabolic consequences of such force generation).

In contrast to the observations made during our length and frequency experiments, however, we found that increases in carbon dioxide concentrations reduced superoxide release from the diaphragm without producing significant changes in force generation by this muscle. The current literature does not provide sufficient information to identify the mechanism by which carbon dioxide exerted this effect; in theory, however, there are several logical possibilities. One possibility is that the activity of the enzymatic pathway responsible for generating superoxide during muscle contraction varies with intracellular pH or carbon dioxide levels. Alternatively, high carbon dioxide concentrations may simply alter superoxide release by muscle or increase entry of this molecule into a number of cellular reaction pathways (e.g., reaction with nitric oxide to form peroxynitrite, etc.) (19). In these latter instances, superoxide release from muscle could be reduced without altering production rate (23).

Regardless of the mechanism by which carbon dioxide affected superoxide release, however, this finding also has important potential pathophysiologic implications. Carbon dioxide levels commonly rise with developing respiratory failure in patients, with blood concentrations often reaching the levels examined in the current report. Our findings would suggest that this physiologic perturbation, taken alone, should act to decrease free radical release from the diaphragm. It is worth noting, moreover, that the development of progressive respiratory failure in patients with many forms of lung disease is commonly associated with simultaneous reductions in inspiratory muscle length, increases in carbon dioxide levels, and gradual reductions in inspiratory muscle force generation (caused by fatigue or reductions in central neural outflow) (24). The present data would suggest that all of these changes could act to reduce diaphragmatic muscle superoxide release, which may, therefore, fall over time in many clinical conditions associated with the gradual development of respiratory failure.

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